

## LESSON 3-4 • SESSION 2 • PRACTICE • FORM A

## Interpreting Points on Ratio Graphs

Learning target – I can explain what a point on a ratio graph means in the situation it describes. • 6.RP.A.3a

**REMEMBER**

- A point  $(x, y)$  says  $x$  of one thing goes with  $y$  of another. **Read the axis labels.**
- The **unit rate point** is  $(1, r)$ . If  $x$  is not 1, divide: for  $(3, 15)$ ,  $15 \div 3 = 5$ , so  $(1, 5)$ .
- To find  $x$  from  $y$ , divide  $y$  by the unit rate: rate 5,  $y = 30$ , so  $x = 6$ .

**WORKED EXAMPLE**

$x$  = pounds of apples,  $y$  = cost in dollars. The line passes through  $(4, 12)$ . Find the cost of 6 pounds.

**Step 1** Read it: 4 pounds cost \$12.

**Step 2** Divide:  $12 \div 4 = 3$ , so the unit rate point is  $(1, 3)$ .

**Step 3** Multiply:  $6 \times 3 = 18$ , so  $(6, 18)$ .

**Answer** \$3 per pound; 6 pounds cost \$18

**WORDS** unit rate point (*punto de tasa unitaria*) • interpret (*interpretar*) • coordinates (*coordenadas*)

**PRACTICE**

show your work • label units

1. Find the unit rate point  $(1, r)$  for a line through each point.

a.  $(3, 15)$  \_\_\_\_\_

b.  $(4, 36)$  \_\_\_\_\_

c.  $(6, 15)$  \_\_\_\_\_

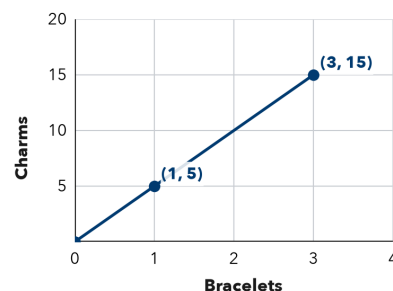
d.  $(5, 40)$  \_\_\_\_\_

2. **Lucy's Charms.** The graph shows bracelets  $(x)$  and charms  $(y)$ .

a. What does  $(0, 0)$  mean? \_\_\_\_\_

b. What does  $(1, 5)$  mean? \_\_\_\_\_

c. Charms for 9 bracelets: \_\_\_\_\_ charms



3. **Water Pump.** A graph of minutes  $(x)$  and gallons  $(y)$  passes through  $(1, 8)$  and  $(5, 40)$ .

**Hint** Divide the gallons by the unit rate.

a. Unit rate: \_\_\_\_\_ gallons per minute

b. What does  $(5, 40)$  mean? \_\_\_\_\_

c. Minutes to pump 72 gallons: \_\_\_\_\_ minutes

4. **Snow Shoveling.** A graph of driveways  $(x)$  and dollars  $(y)$  passes through  $(1, 9)$  and  $(4, 36)$ . What is the pay for 11 driveways?

**Hint** Use the unit rate point  $(1, 9)$ .

Show your work.

**Answer** \_\_\_\_\_ dollars

5. **Drone Flight.** A graph of seconds  $(x)$  and meters  $(y)$  passes through  $(1, 12)$ . What point shows 8 seconds?

Show your work.

**Point** \_\_\_\_\_

6. **Robot Speed.** A graph of seconds (x) and meters (y) passes through (3, 12).

**Hint** Divide 12 by 3 to get the unit rate.

- a. Unit rate point: \_\_\_\_\_  
b. Distance in 10 seconds: \_\_\_\_\_ m

Show your work.

7. **Bike Racks.** A graph of racks (x) and spaces (y) passes through (4, 28). Find the unit rate point. Is (9, 63) on the line?

**Hint** Check: does  $9 \times$  the unit rate equal 63?

Show your work.

Yes or no \_\_\_\_\_

8. **Two Bike Riders.** Both graphs show hours (x) and kilometers (y). Rider 1's line passes through (3, 36). Rider 2's passes through (4, 40).

- a. Each rider's unit rate: \_\_\_\_\_  
b. Whose line is steeper? What does that mean? \_\_\_\_\_  
c. How far ahead is the faster rider after 6 hours? \_\_\_\_\_ km

Show your work.

## FIND THE ERROR

explain the mistake, then fix it

9. **Noah's Work.** A graph of items (x) and cost in dollars (y) passes through (1, 8). Noah says, "This means 8 items cost 1 dollar." Explain his mistake and say what the point really means.

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## TEST PRACTICE

choose the best answer

10. On a ratio graph, the point (1, 9) shows the –  
A origin                      B x-value only                      C unit rate                      D total
11. A ratio graph passes through (5, 20). The unit rate is –  
A 4                      B 15                      C 25                      D 100

## SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|               |         |         |        |         |         |          |              |              |               |
|---------------|---------|---------|--------|---------|---------|----------|--------------|--------------|---------------|
| (1, 0.4)<br>Q | 45<br>R | 40<br>E | 9<br>A | 99<br>P | 12<br>D | 576<br>Z | (8, 96)<br>H | (96, 8)<br>X | (1, 2.5)<br>G |
| #1c           | #2c     | #3c     | #4     | #5      | #6b     | #8c      |              |              |               |

## CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_